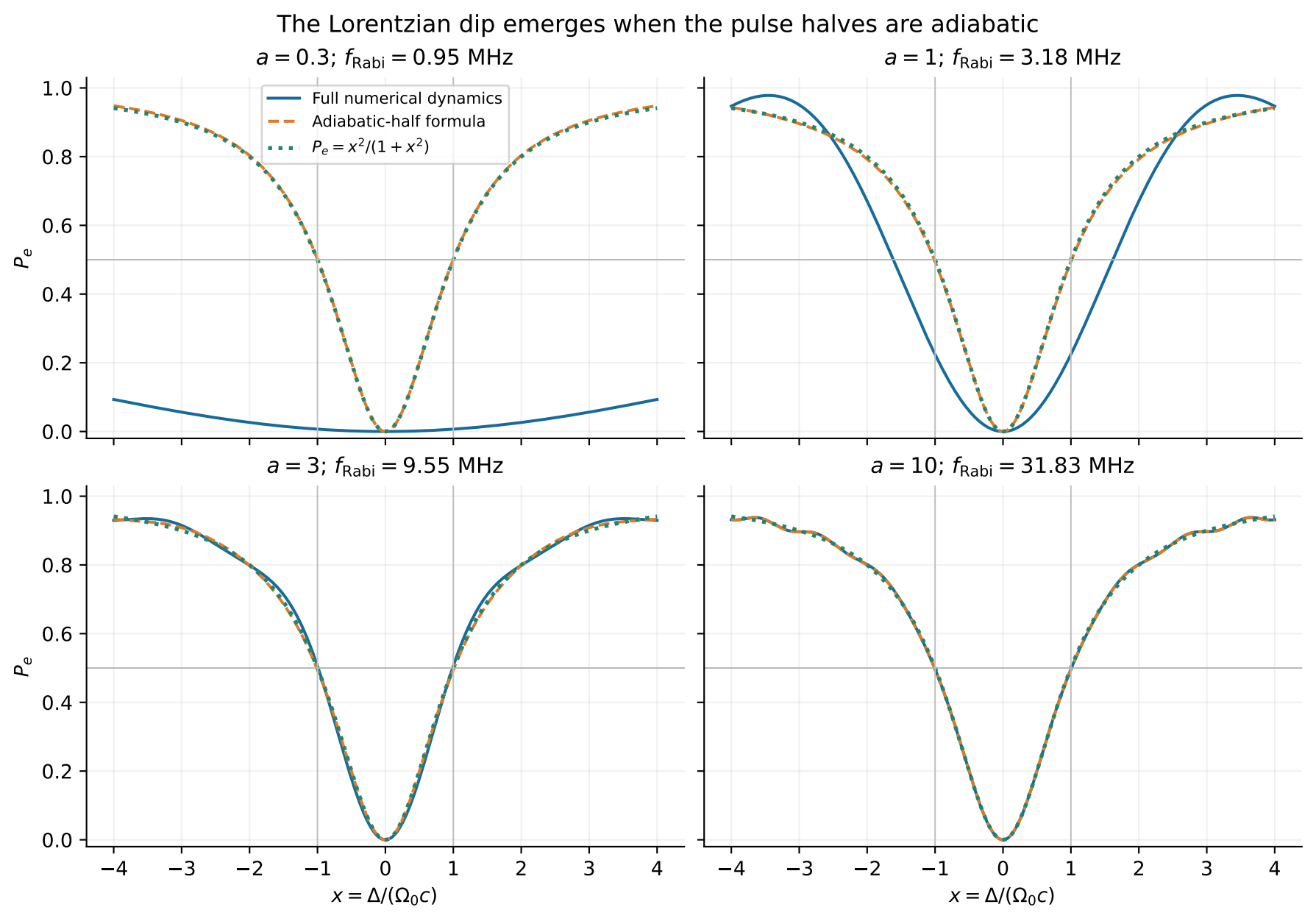
The Lorentzian dip emerges when the pulse halves are adiabatic
$a = 0.3$; $f_{\mathrm{Rabi}} = 0.95$ MHz
$a = 1$; $f_{\mathrm{Rabi}} = 3.18$ MHz
$a = 3$; $f_{\mathrm{Rabi}} = 9.55$ MHz
$a = 10$; $f_{\mathrm{Rabi}} = 31.83$ MHz
Full numerical dynamics
Adiabatic-half formula
$P_e = x^2/(1+x^2)$
$P_e$
$x = \Delta/(\Omega_0 c)$
0.0
0.2
0.4
0.6
0.8
1.0
−4
−3
−2
−1
0
1
2
3
4